Cafe Management System

Product Requirements Document (PRD)

Prepared by: Farha Jabin Oyshee

A platform to streamline daily cafe operations, including product management, order tracking, inventory control, and reporting.

Stakeholders

- **Product Manager**
- **Engineering Lead**
- **Design Lead**
- **Operations Manager**
- **Customer Support Lead**

Status

Draft - Initial PRD for review and refinement.

1. Background

The cafe industry is growing rapidly, with urban cafes playing a vital role in local social and business communities. However, most cafes face challenges in managing operations, inventory, and customer orders due to the absence of streamlined solutions.

The Cafe Management System is designed to address these gaps by providing an intuitive, mobile-first solution tailored for the Bangladeshi market. It enables cafes to efficiently manage their menus, track real-time orders, control inventory, and access actionable business insights.

This system will integrate key local features, such as mobile payment options (bKash, Nagad & more) and offline capabilities, making it highly practical for the region's unique needs.

2. Objective

Cafes in Bangladesh currently rely on manual workflows or generic POS systems that lack local optimizations. The demand for an affordable, scalable system that addresses operational pain points is significant, particularly among independent cafe owners and small chains.

Key differentiators for this system include its ease of use, mobile-first design, and local language support (Bangla). Features like low-stock alerts, real-time order updates, and sales analytics empower owners and managers to run their cafes more efficiently.

3. Assumptions

- Cafe staff prefer systems that are straightforward, mobile-friendly, and localized (e.g., Bangla and English language support).
- Real-time stock alerts and reporting will help reduce wastage and stockouts, boosting profitability.
- The system must work offline for uninterrupted operation in areas with inconsistent internet access.

4. Product

The Cafe Management System consists of six core modules:

1. **Product Entry**: Allows owners to define products, including names, prices, and descriptions.
2. **Category Entry**: Organizes products into logical groups for better menu management.
3. **Outlet Creation**: Supports multi-outlet cafes by enabling unique settings for each location.
4. **Order Management**: Simplifies real-time order tracking and processing for customers and staff.
5. **Inventory Management**: Tracks stock levels, providing alerts for low inventory to prevent stockouts.
6. **Reporting**: Provides business insights through sales and inventory performance reports.

5. Features & Prioritization

Features Ranked by Development Priority

1. **Product Entry**

   Reasoning: Foundational for defining the menu. All other features depend on this.

2. **Category Entry**

   Reasoning: Enhances usability by organizing products into logical groups.

3. **Outlet Creation**

Reasoning: Supports cafes with multiple branches, enabling customized configurations for each location.

4. **Order Management**

Reasoning: Core functionality for daily operations, enabling tracking and status updates for customer orders.

5. **Inventory Management**

   Reasoning: Essential for managing stock levels and ensuring smooth operations.

6. **Reporting**

   Reasoning: Provides business insights once core operational features are functional.

6. User Stories

Product Entry

1. As a Bangladeshi cafe owner, I want to add products with names, prices, and descriptions so I can quickly update my menu to reflect availability.
2. As a manager, I want to list items in Bangla alongside English so local customers understand the menu easily.

Order Management

1. As a cashier, I want to process orders on a mobile device so I can work efficiently in a compact space.
2. As a customer, I want my takeaway order to have an estimated time so I know when to collect it.

Outlet Creation

1. As a franchise owner, I want to track sales across outlets to monitor overall performance.
2. As a manager, I want unique outlet settings for each branch (e.g., pricing variations for Dhaka vs. Chittagong).

Category Entry

1. As a cafe owner, I want to organize products into categories like “Hot Beverages” or “Snacks” for easier navigation.
2. As a manager, I want to quickly switch categories based on promotions or seasonal items (e.g., Ramadan specials).

Inventory Management

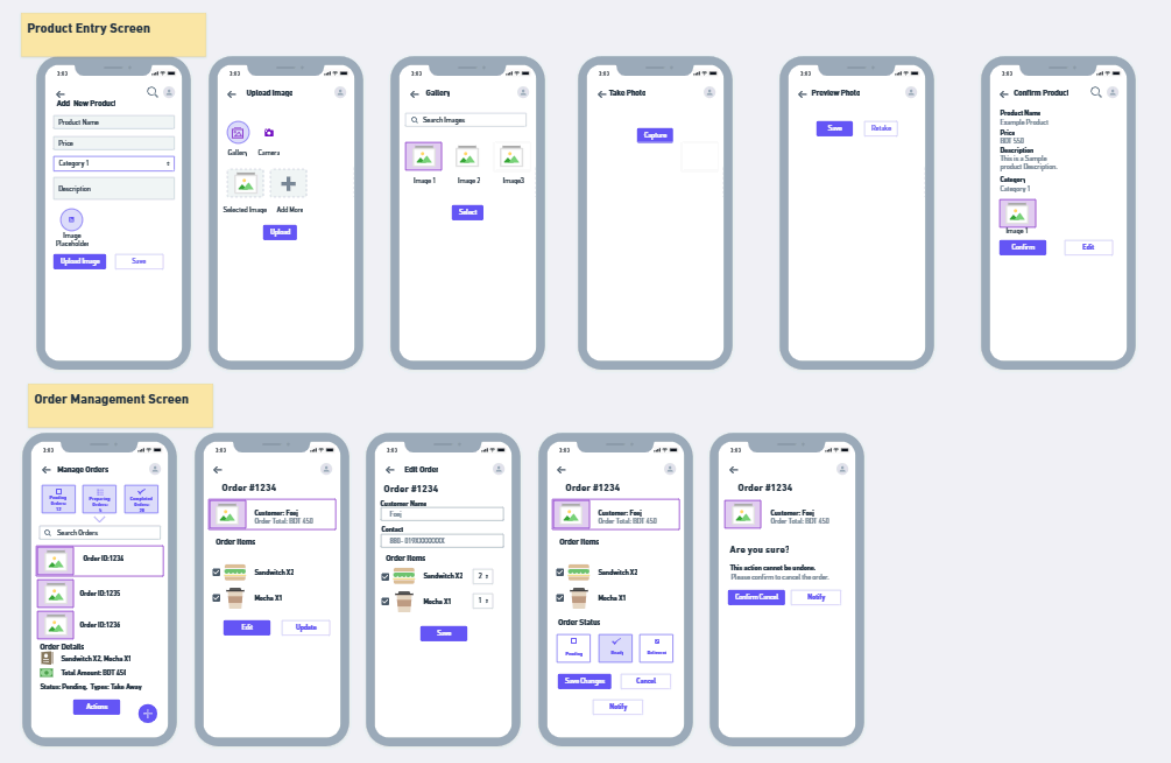
1. As a manager, I want low-stock alerts via SMS so I can act quickly to avoid running out.
2. As an owner, I want inventory reports to ensure suppliers are meeting my needs efficiently.

Reporting

1. As a cafe owner, I want daily sales summaries sent via email so I can stay informed without checking the system.
2. As a manager, I want to identify low-performing menu items to adjust offerings.

7. Wireframes

Link attachment: [Cafe Management Road Map [Wireframe]]



8. Roadmap

Month 1

- Week 1-2: Develop **Product Entry** with dual-language support.
- Week 3: Build a basic **Order Management** module for order processing.

Month 2

- Week 1: Add **Outlet Creation** for cafes with multiple branches.
- Week 2-3: Enhance **Order Management** with takeaway-specific features and SMS notifications.
- Week 4: Introduce **Category Entry** for better product organization.

Month 3

- Week 1-2: Develop **Inventory Management**, focusing on low-stock alerts via mobile.
- Week 3: Implement **Reporting** with daily email summaries and basic dashboards.
- Week 4: Final testing, localization tweaks, and deployment.

Link attachment: [Cafe Management Road Map]

5. Technology Stack

Frontend

- **Vue.js** or **React.js**: Mobile-friendly frameworks with quick loading times, ideal for users on slower internet connections in some areas of Bangladesh.
- **Material-UI** or **Tailwind CSS**: Lightweight and flexible for building responsive UI designs, for intuitive and localized design.

Backend

- **Node.js (Express.js)**: Lightweight and scalable for cafes of varying sizes.
- **Laravel (PHP)**: A widely-used alternative in Bangladesh, supported by local developers.
- **Django** as an alternative if Python expertise is more accessible locally.

Database

- **PostgreSQL**: Structured data for transactions and inventory. Affordable and reliable for small businesses.
- **MongoDB** for dynamic data like inventory movements.
- **Firebase**: Real-time capabilities for order updates in busy cafes.

Hosting:

- Use **AWS Lightsail** or **DigitalOcean** for affordable hosting.

Additional Considerations:

- **Payment Gateway**: Implementation of bKash, Nagad, and SSLCommerz to support local payment options. Integrate advanced features like QR-based payments and wallet functionality to enable seamless, cashless transactions.
- **SMS Alerts**: Leverage SMS providers such as Robi Axiata or Grameenphone for real-time notifications.
- **Cloud Hosting**: Utilize AWS Lightsail or DigitalOcean for scalable and cost-efficient cloud infrastructure tailored to local requirements.

N.B.: *Further implementation will depend on detailed requirements, cost analysis, and budget considerations.*